Supply vessel routing for offshore installations

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DOI: 10.1016/j.example.2024.01.001 (2024)

Abstract. We study supply vessel routing with time windows for platforms on the Norwegian continental shelf using a heuristic.

2 Problem description

The fleet consists of 3 PSVs sailing at 12 knots. Each installation has a service time of 4 h. The average distance is 90 nm. Instances have n = 20 installations.

Table 1 lists the vessels.

Vessel	Speed (kn)	Capacity (t)
PSV-A	12	4500
PSV-B	14	3200

5 Results

The best plan reaches a total cost of 1 234 567 NOK with an optimality gap of 0.8%. Operations stop when Hs exceeds 2.5 m.